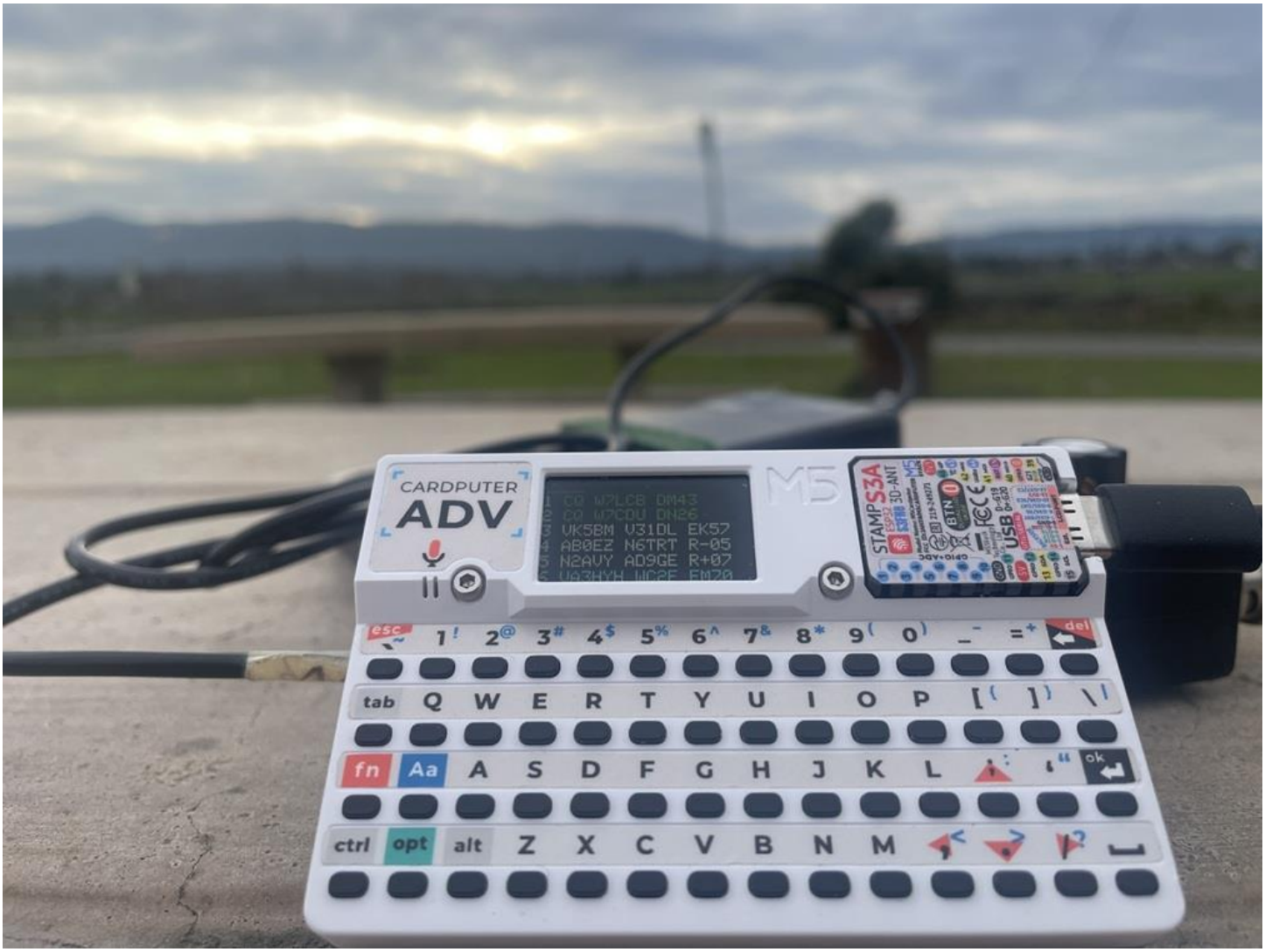
Mini-FT8 User Manual



Mini-FT8 is built on Karlis Goba's ft8_library. It's also a joint adventure between Zhenxing (N6HAN) and Wei (AG6AQ), with inspiration from DXFT8 by Barb (WB2CBA) and Charley (W5BAA). It has been a great learning platform for me, and I hope you find it just as fun to use.

73,

Wei AG6AQ

Revision	Release Date	Notes
V1.4.0	2026/03/02	• Add ARRL Field Day Support (type 0.3/0.4) • Add python script to set accurate time from PC • Rename StationData.ini to Station.ini, will preload from SD card if present • Bug fix: in Tx page, press "1" only toggles the first two messages, now toggles between all messages within the same slot (even or odd)
V1.3.3	2026/02/12	• Tx scheduler works more efficiently
V1.3.2	2026/01/29	• Add SD card support

		• Add a startup screen
V1.3.1	2026/01/28	• Fix hashtable overflow
V1.3	2026/01/24	• Add support for nonstandard callsign • Hook up SkipTX1 to autoseq

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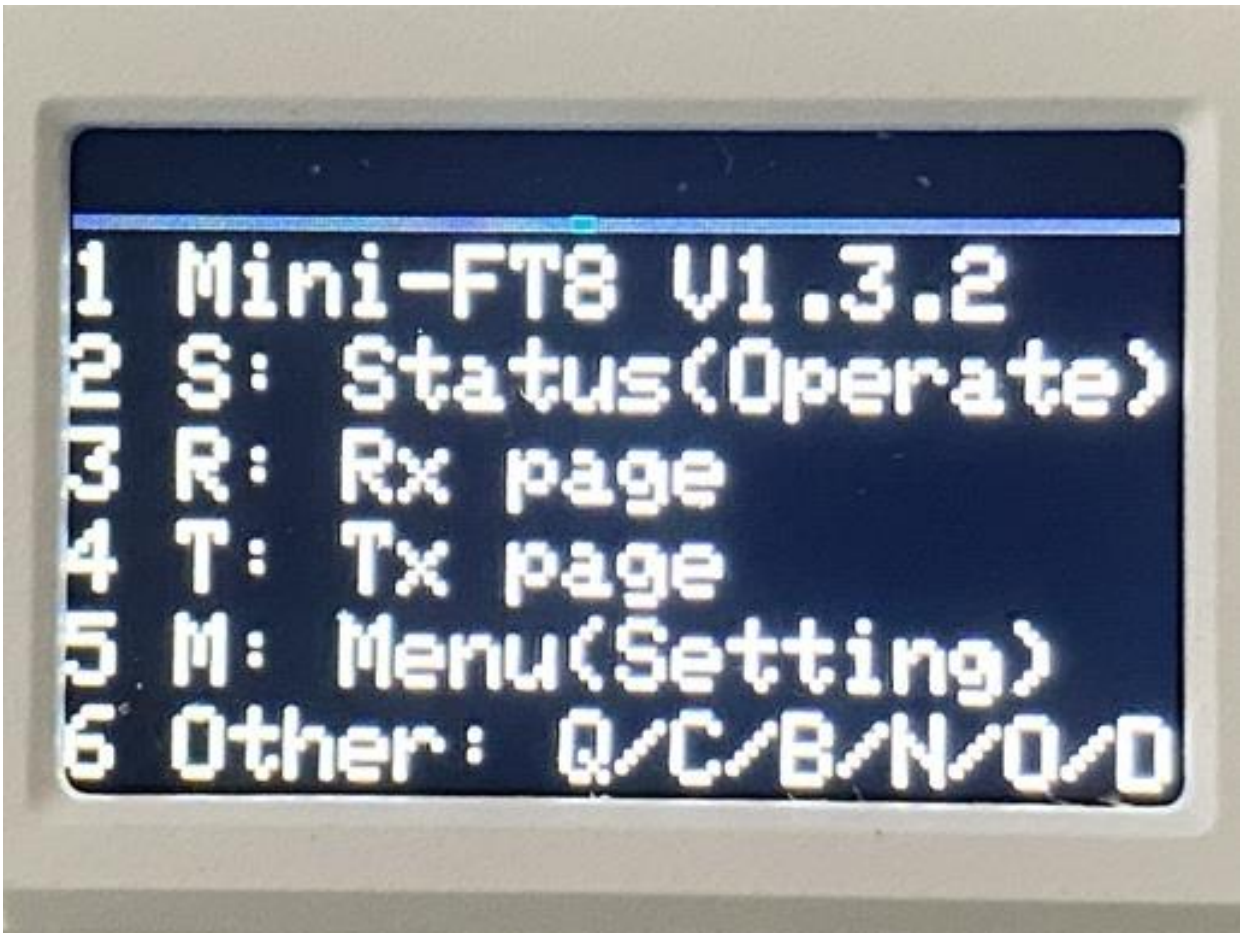
1. Screen Layout



(Startup Screen)

The Cardputer ADV has a tiny 1.14" 240×135 TFT screen. It has obvious drawbacks, but for our POTA/SOTA use cases we don't stare at the screen much. The small display also helps reduce power consumption. The built-in 1700 mAh battery lasts for hours (10+ hours in my testing).

- Top 18 px: Waterfall (shows band activity / conditions) (200-3000Hz)
- Next 3 px: 15s countdown bar (a tiny 5x3 blue dot shows Cursor)
- Main area 114 px (19×6): 6 lines of text — this is the primary operating area

There are two simple ways to improve readability if needed:

- Use a 3× plastic magnifier (same size as Cardputer)
- Use a smartphone camera and zoom in

2. UI System

Mini-FT8 has a simple UI that can be adapted to a keyboard, touch screen, buttons, or remote control. When you click a line button (1–6 in our case), each item behaves as one of five types:

1. Select: choose from predefined options (e.g., CQ Type)
2. Action: trigger an action (e.g., Send FreeText)
3. Edit: edit the value in place (e.g., Callsign)
4. Long Edit: open a full-screen editor (e.g., Comment)
5. Information: read-only display (e.g., Battery)

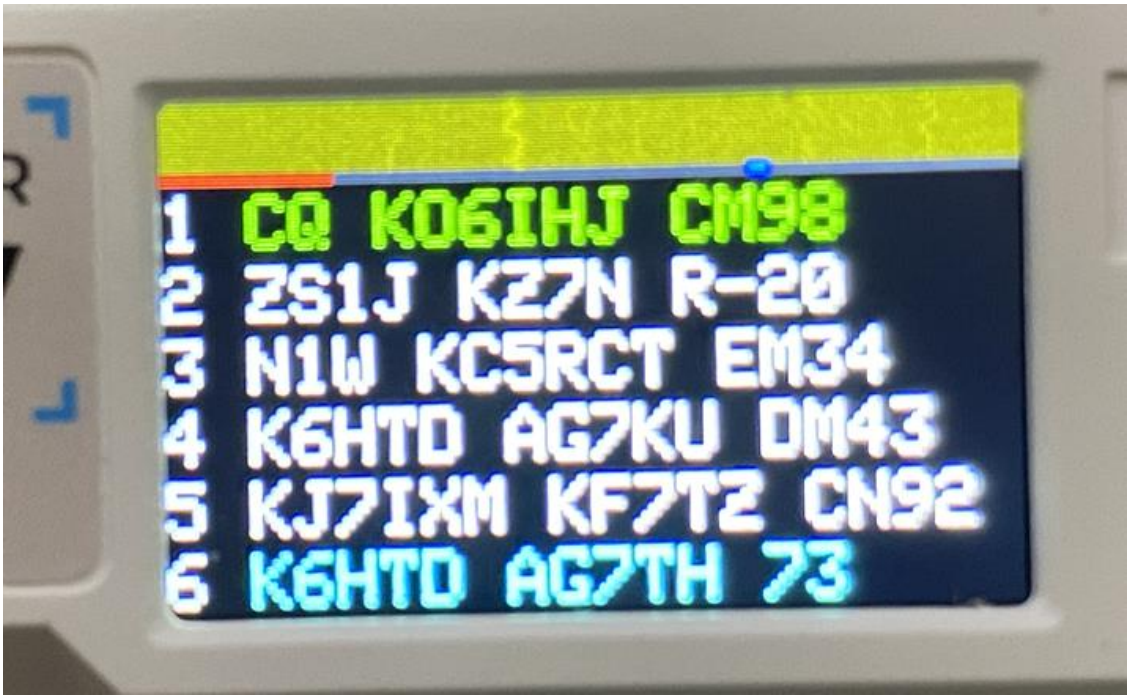
You can switch between pages by clicking:

- **R(x):** Shows decoded messages (default page).
- **S(tatus):** Think of this as a frequently used "quick menu."
- **T(x):** Actively manage the transmission (TX) queue.
- **M(enu):** Settings (3 pages).
- **Q(so):** Shows QSOs from `YYYYMMDD.adi`.
- **B(and):** Change band frequencies (just in case).
- **C(onnect to PC):** Connect to a PC to read/delete files.

Exception: If you're in C page, you must click C again (or type "exit" on the PC) to leave C page.

(There are some zombie pages, e.g. H, L will be removed later)

2.1 Rx: Shows decoded messages (default page)



Mini-FT8 can decode 20+ messages under optimal conditions. Messages are sorted in this order: **reply-to-me**, **CQ**, then **others**.

Only **12 messages** are shown on the **Rx** page. If **line 6** is **cyan**, a **second page** is available—press **Down (.)** to view it (otherwise, don't bother). You can find the **full decode list** in **RxTxLog.txt**.

You can click on one or more **CQ** messages to start QSOs. If the click happens within **4 seconds** of the slot boundary, **Tx starts immediately**; otherwise, it will be **scheduled for the next appropriate slot**.

You can also click **other** messages—this is mainly intended for **stations that called CQ previously**, or a **rare callsign**.

Reply-to-me messages are handled automatically by **auto sequencing**.

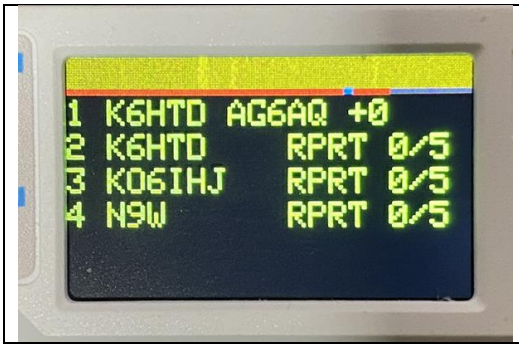
If you're calling CQ or replying to a CQ, Mini-FT8 will automatically switch to the Tx page while transmitting. This is a feature, not a bug.

2.2 Status: Think of this as a frequently used "quick menu."

1. Select: [OFF|EVEN|ODD]
2. Action: Connect/Sync to QMX
3. Select: Select band from active band (Menu: N3)
4. Select: Turn tune ON/OFF
5. Edit: Set date
6. Edit: Set time

Tip: setting time (RTC sync trick)
Example: if it's 04:08:05 now, give yourself ~10 seconds. Enter 040814. When your watch reaches 04:08:13, press OK. The RTC will land close to the correct second, and auto-sync will use decoded FT8 messages to adjust RTC.

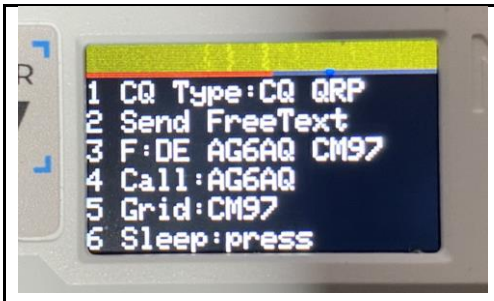
2.3 Tx: Actively manage the transmission (TX) queue



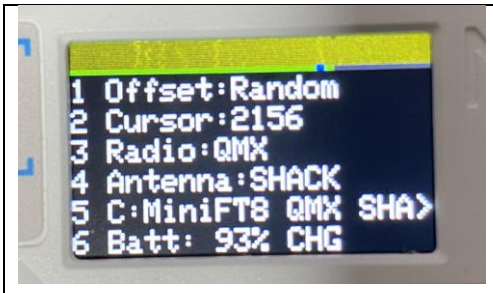
- Line 1: Next message that is scheduled for transmission. Press 1 to cycle through queued messages within the same slot (even or odd) as Line 1
- Lines 2–6: Queued messages. Press 2–6 to delete the corresponding entry.
- You can press Esc to cancel an active transmission. The message remains in the queue until it is deleted or expires.
- The queue supports up to 9 messages.

2.4 Menu: Settings (3 pages)

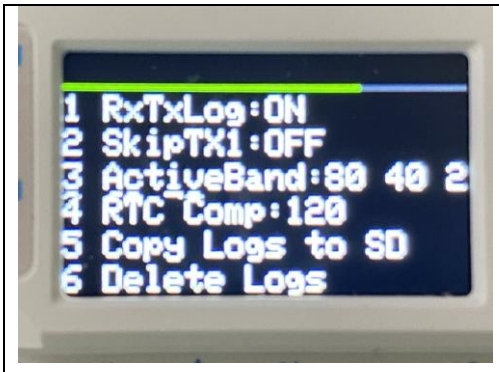
Page 1 (M)

	1. Select: [CQ | SOTA | POTA | QRP | FD | FreeText] 2. Action: Send FreeText in the next slot 3. Long Edit: Open FreeText editor (full edit) 4. Edit: Up to 11 characters callsign 5. Edit: 4-charater Grid 6. Action: Put Cardputer into sleep (Use BtnGo to wake it up) **FreeText – intended uses:** • Plain free text, e.g. "TNX 73" • Typed like a standard CQ, e.g. CQ EU W6XYZ CM97 If it matches a standard FT8 message format, it will be encoded/sent as a standard message (not FreeText). • Self-spot for sotamat, e.g. "SM W6ABC/PKW7" • Set field day exchange, see 5.4 Operating Mini-FT8 for ARRL Field Day

Page 2 (M. or N)

	1. Select: [Random | Cursor | Rx] • Random: choose a Tx offset within 500–2500 Hz. • Cursor: Use the current cursor frequency for Tx offset. • Rx: Use the currently selected Rx message frequency (the one you clicked) as a Tx offset. 2. Edit: Tx offset (only used when Select = Cursor) • a tiny blue dot shows the cursor on countdown bar, if you're not using it and want to hide the dot, enter a value <200 or >3000) 3. Select: QMX (only radio supported right now) 4. Edit: just a short note for yourself 5. Long Edit: Comment saved into the .adi file • In the ADIF comment field, /Radio will be replaced by item 3 (Radio selection), and /Ant will be replaced by item 4 (Antenna). 6. Information

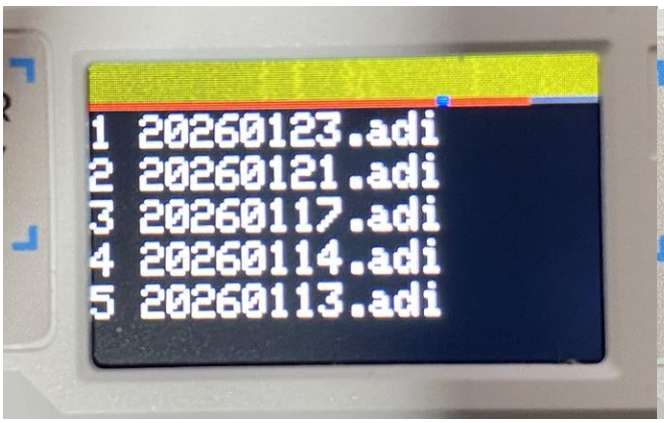
Page 3 (M.. or O)



1. Select: [ON | OFF] Enable/disable RxTxLog.txt logging
2. Select: [ON | OFF] When replying to a CQ, skip the grid exchange and reply with signal report instead
3. Long Edit: Band selection (choose from 12 bands, in B pages)
4. Edit: Hardware RTC compensation (calibrates clock drift)
   - more details see: RTC Deep Sleep Compensation
5. Copy all logs and Station.ini to SD card
6. Delete all logs (except Station.ini) on device!!!

2.5 Qso: Shows QSOs from YYYYMMDD.adi

Informative, use 1-6, Up/Down and Esc to navigate different .adi files

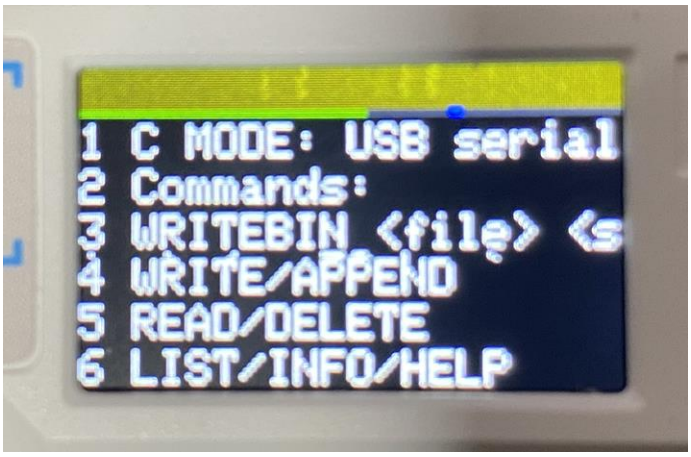


2.6 Band: Change band frequencies (just in case)

Edit band frequencies (Most people don't need to edit these pages)

2.7 Connect to PC: Read/Delete files

Information (Help Info for PC Terminal)



3. Basic Operations

Startup Screen:

- Mini-FT8 V1.3.2
- S – Status (Operate)
- R - Rx page
- T - Tx page
- M – Menu (Setting)
- Other: Q/C/B/N/O/D

1. Power on **Mini-FT8** and double-check the **Menu** settings are correct.
2. Press S → go to the **Status** page.
3. Press **5/6** → Set **Time/Date (UTC)**.
4. Press **3** → select band *(Active band is configured in **Menu page 3**)*.
5. Press **2** → **Connect to QMX**.
    - Then **power on the QMX**
6. Press **2** → **Sync to QMX**.
    - If you don't see **"Sync to QMX"**, it means **there is no connection**.
7. Press **4** → tune the antenna (if needed).
8. Press **1** → enable **Beacon** (Even or Odd slot) if you're doing **POTA/SOTA** activation.
9. Go to the **Rx/Tx** page and have fun.

4. File System and Logging

Mini-FT8 uses on-chip flash (4 MB SPIFFS) to store settings and logs.

4.1 Settings: /spiffs/Station.ini

Saves device settings. You can download/upload it, but it's generally best to leave it alone. If an SD card is inserted with Station.ini, it will be copied over to spiffs.

4.2 RX/TX Log: /spiffs/RxTxLog.txt

This log serves three purposes (enable/disable in Menu → Page 2):

- Debug AutoSeq behavior
- Save received messages for propagation evaluation
- Help confirm QSOs that failed at signoff (a minimal QSO only requires callsign exchange)

Maintenance tip: RxTxLog.txt can grow large over time. It's a good idea to download it periodically and delete the copy on the Cardputer.

4.3 ADIF Log: /spiffs/YYYYMMDD.adi

Appended when a QSO is completed (RR73 sent/received).

The QSO viewer (Q) reads this file and shows entries like: hh:mm band call.

4.4 Download Logs

4.4.1 Use SD Card

- Insert a FAT/FAT32 formatted SD card.
- Go to Menu page 3 (O), press 5 — all log files will be copied to the SD card.
- If you use M5Launcher, you can also rename and delete files on the SD card directly.

4.4.2 Use pc_terminal.py

- On M5 Carputer, click C to enter communication
- On PC: *python .\pc_terminal.py COM11* for multiple commands
- On PC: *python .\pc_terminal.py COM11 read 20260113.adi* for single command

(files will be written into current directory)

Make sure pyserial is installed (pip install pyserial).

Ubuntu/Linux: use the device node, e.g. python3 pc_terminal.py /dev/ttyACM0 (or /dev/ttyUSB0)

On Ubuntu, ensure your user can access the port: sudo usermod -a -G dialout $USER(then log out/in), or temporarily sudo chmod a+rw /dev/ttyACM0

4.4.3 Use a Serial Port Terminal

The protocol is plain UART, so almost any serial terminal will work.

Examples:

- PuTTY
  - Terminal → Implicit CR in every LF: enable
  - Terminal → Local echo: Force on
- Arduino IDE Serial Monitor
  - Works fine as a simple UART terminal as well.

5. FAQ

5.1 How to Upload Firmware (FW) to Cardputer

ChatGPT is the best resource to get this done.

5.1.1 Method 1: Flash from source (ESP-IDF)

- Install ESP-IDF (matching the project version).
- Connect Cardputer to PC with a USB-C data cable.
- In the project folder:
  - idf.py build
  - idf.py -p <PORT> flash
- Windows port looks like COM11
- Linux port looks like /dev/ttyACM0 (make sure you have permission)

Benefit of this method: you can get the latest firmware without waiting for an official release. Updates are pushed to GitHub after they pass basic sanity checks.

5.1.2 Method 2: Flash released binaries (no build)

Download the release package (bins + scripts).

- Install esptool: pip install esptool
- Run the provided flash script:
  - Windows: flash.ps1 (enter port and filename when prompted)
  - Ubuntu/Linux: flash.sh (enter port and filename when prompted)

```
ESP-IDF 5.5 PowerShell

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----         1/29/2026   7:55 PM           1265 flash.ps1
-a----         1/29/2026   8:00 PM           1553 flash.sh
-a----         1/29/2026   7:04 AM         792315 Mini-FT8 User Manual.pdf
-a----         1/29/2026   7:50 PM         984880 MiniFT8_V1.3.2.bin
-a----         1/18/2026   8:07 AM           5290 pc_terminal.py
-a----         1/15/2026   8:38 PM            501 readme.Ubuntu.txt


PS C:\MiniFT8-idf\mini_ft8_min\release> .\flash.ps1
Mini-FT8 flasher (merged .bin)

Enter serial port (e.g., COM11): COM11
Enter merged firmware .bin path (e.g., MiniFT8_V1.3.2.bin): MiniFT8_V1.3.2.bin

Flashing esp32s3 on COM11 @ 460800...
  File: C:\MiniFT8-idf\mini_ft8_min\release\MiniFT8_V1.3.2.bin
esptool.py v4.11.dev1
Serial port COM11
Connecting...
Chip is ESP32-S3 (QFN56) (revision v0.2)
Features: WiFi, BLE, Embedded Flash 8MB (GD)
Crystal is 40MHz
USB mode: USB-Serial/JTAG
MAC: 10:51:db:55:71:e4
Uploading stub...
Running stub...
Stub running...
Changing baud rate to 460800
Changed.
Configuring flash size...
Flash will be erased from 0x00000000 to 0x000f0fff...
SHA digest in image updated
Compressed 984880 bytes to 529392...
Wrote 984880 bytes (529392 compressed) at 0x00000000 in 6.4 seconds (effective 1231.4 kbit/s)...
Hash of data verified.

Leaving...
Hard resetting via RTS pin...
PS C:\MiniFT8-idf\mini_ft8_min\release>
```

Tip: Use a short, good-quality USB-C cable. Some “charge-only” cables won’t work for flashing.

5.1.3 Method 3: Install Mini-FT8 with M5Launcher

This is the recommended method.

1. install M5Launcher from https://bmorcelli.github.io/Launcher/webflasher.html
2. Copy MiniFT8_V1.x.x.bin directly to SD card
3. Install MiniFT8 FW under M5Launcher menu

If you build binary yourself, you can generate MiniFT8_V1.x.x.bin with:

esptool.py --chip esp32s3 merge_bin --output MiniFT8_V1.x.x.bin --flash_mode dio --flash_size 8MB --flash_freq 80m 0x0 bootloader.bin 0x8000 partition-table.bin 0x10000 mini_ft8.bin

You can put multiple Mini_FT8 revisions on the SD card and use the same card to store log files.

5.2 How to Charge the Cardputer

5.2.1 Charge via USB-C using a normal 5V USB supply.

Leave the Cardputer's power switch ON and use Sleep (Menu M6) when you're not using it. In some cases, you may need to reset the M5 Cardputer to return the USB-C port to serial mode, so VBUS is reconnected to the 5V power rail.

It's easier to use **M5Launcher's charge mode**.

5.2.2 Charge via PORTA

You can also charge the Cardputer via PORTA.

A simple DIY method is to homebrew a charge cable: cut a USB cable you don't mind sacrificing, then connect VBUS (5V) and GND to the 4-pin Grove connector.

- USB VBUS (5V) → Grove VCC
- USB GND → Grove GND

Before plugging in, make sure the micro switch next to PORTA is set to the 5VIN position.

5.3 How to Operate on Frequencies Other Than the "Standard" Ones

If you want to operate on a different dial frequency (for example, 7040 kHz), you can change it on the B(and) pages.

If you want to make 7040 kHz a selectable preset option, you'll need to hack it. A simple approach is to reuse an unused band entry:

- Pick an unused band (for example, 160 m) and change its frequency to 7040 kHz on the B(and) page.
- Then enable that "band" by adding 160 to the Active Band list (Menu → Page 3).

5.4 Operating Mini-FT8 for ARRL Field Day

ARRL Field Day is the most popular on-the-air event, held each year on the fourth weekend of June. Since Mini-FT8 is designed for portable operation, it includes native Field Day message support (FT8 types 0.3/0.4). To use Field Day mode:

- Set CQType to "CQ FD" (Menu M1).
- **Set FreeText to your Field Day exchange**, for example "1B SCV"(Menu M3)

Field Day does not use RST or grid exchange. To save 15 seconds when replying to a CQ FD station, set SkipTX1 = ON so Mini-FT8 sends your Field Day exchange immediately.

A **corresponding fieldday.log (Cabrillo format)** will be saved and copied to the SD card along with the other log files.

Turning on RxTxLog.txt is highly recommended. It provides a complete record of transmitted and received messages so you can recover a QSO if something goes wrong. If everything looks good, you can delete or ignore it afterward.

Sample FD QSO:

CQ FD N6HAN CM87

N6HAN AG6AQ 1B SCV

AG6AQ N6HAN R 1A ORG

N6HAN AG6AQ RR73

AG6AQ N6HAN 73

Sample fieldday.log:

START-OF-LOG: 3.0

CREATED-BY: Mini-FT8

CONTEST: ARRL-FIELD-DAY

CALLSIGN: AG6AQ

CATEGORY-OPERATOR: SINGLE-OP

CATEGORY-TRANSMITTER: ONE

CATEGORY-ASSISTED: NON-ASSISTED

CATEGORY-BAND: ALL

CATEGORY-MODE: MIXED

CATEGORY-POWER: LOW

CATEGORY-STATION: PORTABLE

LOCATION: SCV

OPERATORS: AG6AQ

QSO: 7074 DG 2026-03-01 1948 AG6AQ 1B SCV N6HAN 1A ORG

END-OF-LOG:

5.5 How to set date/time from a PC

You can set date/time from PC just before you on-the-air session.

- Download set_rtc_sleep.py from: https://github.com/wcheng95/Mini-FT8/tree/main/tools
- On the M5 Cardputer, press C to enter communication mode.

- On the PC, run: python .\set_rtc_sleep.py

If the script fails to put the M5 into sleep, use Menu M6 instead (use “BtnGo” to wake it up). You can also simply leave the M5 on to keep date/time more accurate - battery lasts a few hours.

6. ToDo List and Known Bugs